higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

QUANTITY SURVEYING N6

15 APRIL 2019

This marking guideline consists of 10 pages.

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QUESTION 1

1.1
1.1.1 Numbers (no.)
1.1.2 Metres (m)
1.1.3 Metres (m)
1.1.4 Square metres (m²)
1.1.5 Numbers (no.)
1.1.6 Square metres (m²)
1.1.7 Square metres (m²)

(7 × 1) (7)

1.2 1.2.1 Site clearance refers to the preparation of the site✓ by removing all rubbish, small trees and bushes✓ before setting out the works.

1.2.2 Working space is the allowance for additional excavations✓ beyond the original excavations to create space✓ for workmen to execute their work.

(2 × 2) (4)

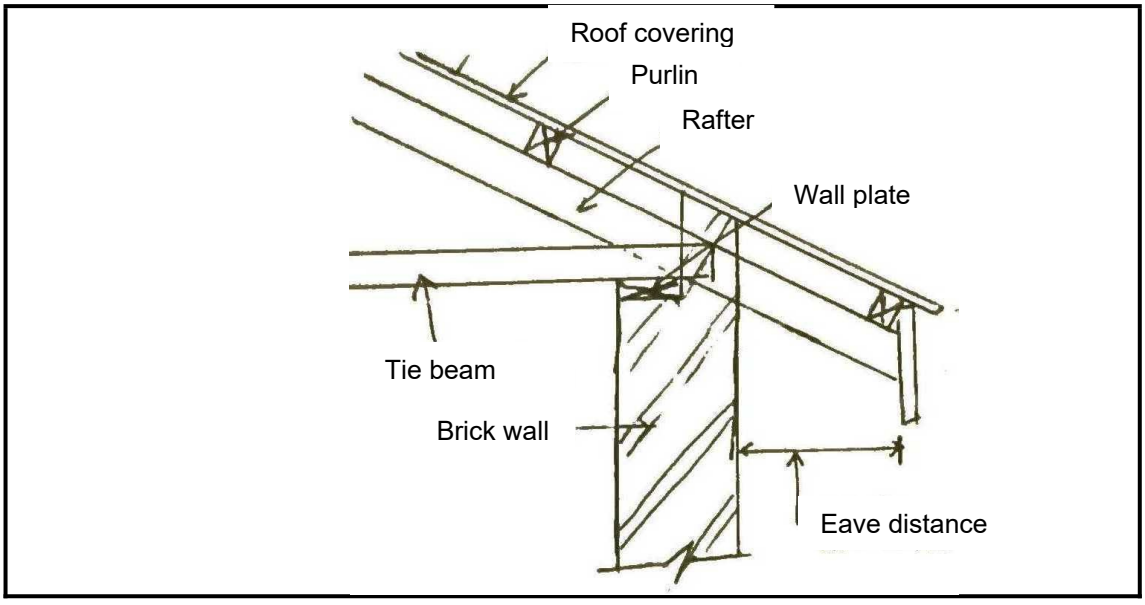
1.3 1.3.1 This process consists of three stages: taking-off;✓ working-up✓ and billing.✓

1.3.2 This process consists of taking-off.✓ Measurements of each item are entered on cut-and-shuffle slips for billing.✓ Abstracting is eliminated.✓

1.3.3 Special computer software is developed.✓ Taking-off procedure is done to the extent required by the program used and fed into the program.✓ A complete bill of quantities can be printed immediately.✓

(3 × 3) (9)

1.4



(5)
[25]

QUESTION 2

	19,00 15,50	✓	C.s. & prep. site for bldrs. wrk.✓ 1 000 – 350 = 650 ÷ 2 = 0,325 m 15,000 × 11,500 2 x 2 m 4,000 4,000 19,000 × 15,500		53,00 0,50	✓	Exc. back from bsmnt. Hole for w.s. n.e. 500 m dp. ✓
					53,00 1,00	✓	Ditto, but exc. 500 mm n.e. 1,5 m dp. ✓
					53,00 1,50	✓	Ditto, but exc. 1,5 m n.e. 3 m dp. ✓
	15,00 11,50 2,00	✓	Excav. in ord. earth for bsmnt. Hole n.e. 2 m dp. ✓		53,00 0,30	✓	Ditto, but exc. 3 m n.e. 4,5 m dp.✓ 3,300 - 3,000 = 0,300 m
			Ditto, but exc. 2 m n.e. 4 m dp. ✓ 2 950 + 350 = 3 300 – 2 000 = 1,300 m		53,00 3,60	✓	R.o.c to sides of bsmnt. Hole exc. 1,5 m dp. from g.l. ✓ 3,300 + 0,300 = 3,600 m ✓
	15,00 11,50 1,30	✓					
	49,00 1,00 0,30	✓	Excav. for bsmnt. trench exc. 2 m n.e. 4 m from g.l. ✓ 2/15,000 = 30,000 2/11,500 = 23,000 53,000 Less 4/ 1,000 4,000 49,000 m ✓		45,00 0,30	✓	Ditto,but to int. sides of bsmnt. trench exc. 1,5 m dp. from g.l.✓ 53,00 - 8/1,000 = 45,000 m✓
					49,00 1,00 0,30	✓	25 MPa mass conc. in bsmnt. ftg. ✓
			(1)				(2) **[25]**

	49,00 0,50	✓	350 mm thk. bk. bsmnt. wl.✓ 0,350 + 0,150 = 0,500 m		13,65 10,15 0,15	✓	H.c. in bsmnt flr. ✓ 15,000 × 11,500 -2/675 1,350 1,350 13,650 × 10,150 m
	51,48 3,05	✓	110 mm t hk. bsmnt. Skin wl. ✓ 2,950 + 0,250 - 0,150 = 3,050 m 53,000 -4/325 + 55 = 1,520 51,480 m ✓		13,21 10,59 0,15	✓	25 MPa mass conc. in bsmnt. flr. incl. s.o.c. hor. top surf. ✓ 13,650 × 10,150 +2/0220 0,440 0,440 13,210 × 10,590
	50,74 2,73	✓	220 mm thk. bk. bsmnt. wl. ✓ 3,050 - (2/0,150 + 0,020) = 2,730 m 51,480 - 4(55 + 20 + 110) = 50,740 m ✓		14,09 10,59	✓	20 mm thk. hor. mastic asphalt.✓ Length = 15,000 - 2(325 + 130) = 14,090 m Width = 11,500 - 2(325 + 130) 10,590 m
					51,22 3,05	✓	Ditto, but vert. ✓ 51,480 - 4(65) = 51,220 m
	51,70 0,33 3,30	✓	Ext. b/fl. To sides of bsmnt. wl. ✓ 53,000 - 4/0,325 = 51,700 m 2,950 + 0,350 = 3,300 m	2 2	14,09 10,59	✓ ✓	Extra for triangular mastic asphalt fillet. ✓
	47,65 0,33 0,35	✓	Int. b.fl. to sides of 250 mm thk. bk. wl. ✓ 49,000 – 4/338 = 47,645 m ✓				
			(3)				(4)

QUESTION 3

3.1

2							
	0,30		25 MPa reinf. conc. in				
2	0,30	✓	cols. ✓				
II	3,50		-B1;B3				
	0,15						
	0,15	✓					
2	3,50		-A2;B2				
	0,50						
	0,30	✓	2/0,400 = 0,800				
	3,50		Less 0,300				
			-A1;A3 0,500 m ✓				
2							
4							
	0,30		Frmwrk. to sides of cols.✓				
	3,50	✓	B1; B3				
	2	✓	Ditto, but to sides of circular cols. ✓				
			-A2; B2				
2							
4							
	0,40		Ditto, but to l-shaped				
	3,50	✓	cols.✓				
			-A1; A3				
			(End of QUESTION 3.1)				
			(1)				(2)

(11)

3.2

	9,50 8,00 0,15	✓	25 MPa reinf. conc in slab incl. s.o. & c. hor. top surf. ✓ 7,300 + 0,300 + 0,400 = 8,000 m 2/4,300 + 3/0,300 = 9,500 m	2	6,70 4,30	✓	Frmwrk. to soffits of slab. ✓ 7,300 – 2/0,300 = 6,700 m (4)

3.3

	33,80 0,30 0,35	✓	25 MPa reinf. conc. in bms. ✓ 2/8,000 = 16,000 2/9,500 = 19,000 35,000 Less 4/0,300 1,200 33,800 m ✓ 0,500 – 0,150 = 0,350 m				
	8,00 0,30 0,30	✓					
	33,80 0,30	✓	Frmwrk. to soffits of bms. ✓				
	8,00 0,30	✓					
2	33,80		Frmwrk. to sides of bms. ✓				
2	0,35	✓					
	8,00 0,30	✓					
	35,80 0,15	✓					(10) **[25]**

QUESTION 4.1 **EXAMINATION NUMBER:** | | | | | | | | | | | | | |

2 ✓	✓		75 x 110 mm meranti tbr. once rebated.		5,26		Priming to backs of frame.
✓	1,53	3.06					
✓	1,10	1,10	-Stiles -Head				2 x 1,099 = 2,198 2 x 1,530 = 3,060 5,258 m
	✓	4,16	✓ ✓				
2							
	1,53		Two cts. clear vanish to tbr. surfaces.				
	0,26	0,80					
	1,10		-Stiles Girth		2	2	300 x 472 mm meranti tbr. fanlight w. two 50 x 44 mm stiles, once rebated, one 50 x 44 mm top & one bot rail, once rebated.
	0,26	0,29	2 x 75 = 0,150				
			-Head 0,110				
		1,09	0,260 m				
			75 x 110 mm meranti tbr. twice rebated.	2			
				2			Two cts. clear vanish to flat surfaces of fanlight.
	1,53	1,53	-Mullion		0,47		
	1,10	1,10	-Transome		0,30		
		2,63					
			Two cts. clear varnish to exposed tbr. surfaces.				
	1,53				1	1	472 x 1 085 mm meranti tbr.casement w. two 50 x 44 mm stiles, once rebated, one 50 x 44 mm top & one bot. rail, once rebated.
	0,37	0,57					
	1,10						
	0,37	0,41	Girth				
			2 x 75 = 0,150				
		0,98	2 x 110 = 0,220				
			0,370 m				
				2			
			75 x 200 mm meranti tbr. twice rebated & once weathered.		0,47		Two cts. clear varnish to flat surfaces of casement.
	1,10	1,10	-cill		1,09		
			Two cts. clear varnish to exposed tbr. surfaces.				
	1,10						
	0,35	0,39	Girth				
			2 x 75 = 0,150				
			0,200				
			0,350 m				
			(1)				(2)

(7)

QUESTION 4.2 **EXAMINATION NUMBER:** | | | | | | | | | | | | | |

				TIMBER CASEMENT WINDOW ✓					
				No					
Carpentry & Joinery		✓							
				300 x 472 mm mer.			tbr. f.l. w. 50		
✓				x 44 mm stiles,			Top & bot.		
M		✓		rails once reb.					
75 x 110 mm mer. tbr. once reb.				2	2				
14,16	1	✓							
				2 No					
✓									
14M		✓							
				No.					
✓		✓							
				472 x 1 085 mm			Mer. tbr. cmt.		
M				w. 50 x 44 mm stls once reb.			Top & bot. rls.		
75 x 110 mer. tbr.		twice reb.		1	2				
2,63	1			1 No.					
3M									
M									
75 x 200 mm mer.		tbr. twice							
reb. once wtrd.									
1,10	1								
1M									

(9)

QUESTION 4.3

	TIMBER CASEMENT WINDOW ✓				
	Bill No. 1 ✓				
✓	Carpentry and Joinery ✓	✓	✓		
1.	75 × 110 mm meranti timber once rebated ✓	M	14		
2.	75 × 110 mm meranti timber twice rebated	M	3		
3.	75 × 200 mm meranti timber twice rebated and once weathered	M	1		
4.	300 × 472 mm timber fanlight with 50 x 44 mm stiles, top and bottom rails once rebated	No	2		
5.	472 x 1 085 mm timber casement with 50 x 44 mm stiles, top and bottom rails once rebated	No	1		
	Carried to summary ✓			R	✓

(9)

WORKING-UP MARK ALLOCATION FOR QUESTION 4

Squaring		
Check dimensioning. (ticks)	2	
Square dimensions.	1	
Check squaring answers. (ticks)	1	
Check final answers.	1	
Transfers to abstract.	1	
Check transfers.	1	(7)
Abstracting		
Section of work	1	
Trade	1	
Units	1	
Page referencing	1	
Use of columns	2	
Reducing	1	
Transfers to bill	2	(9)
Billing		
Section of work	1	
Bill no.	1	
Trade	1	
Item no.	1	
Descriptions (No abbreviations)	1	
Units	1	
Quantities	1	(9
Summary	2	)

[25]

TOTAL: 100

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